

BENJAMIN HOFFMAN

bhoffman@ethz.ch | LinkedIn | Personal Website | Swiss & American

EDUCATION

ETH ZURICH PhD Researcher in the Networked Systems Group ‣ Current research: machine learning and control for large scale systems ‣ Supervised by Prof. Laurent Vanbever	Zurich, Switzerland Feb. 2025 - present
ETH ZURICH MSc in Robotics, Systems and Control — GPA: 5.7/6.0 ‣ Focus on control and machine learning	Zurich, Switzerland Sep. 2022 - Dec. 2024
LUND UNIVERSITY Academic Exchange Semester — GPA: 5.7/6.0 ‣ Focus on control and machine learning	Lund, Sweden Jan. 2022 - Jun. 2022
ETH ZURICH BSc in Mechanical Engineering — GPA: 5.2/6.0 ‣ Focus on mechatronics and control	Zurich, Switzerland Sep. 2018 - Aug. 2021
KANTONSSCHULE WETTINGEN Swiss Matura and IB Diploma — GPA: 5.7/6.0 (top 0.5%, salutatorian) ‣ Major in law and economics, academic exchange semester at the Gymnase de Nyon (VD, Switzerland)	Wettingen, Switzerland Aug. 2014 - Jun. 2018

RESEARCH EXPERIENCE

NETWORKED SYSTEMS GROUP — ETH ZURICH Student Research Assistant — part time ‣ Implemented a scalable real-world data collection and algorithm evaluation platform using Python and Amazon MTurk ‣ Performed distributed testing of ML-based streaming algorithms to evaluate their performance in real-world conditions	Zurich, Switzerland May 2024 - Dec. 2024
MASTER THESIS — ETH AI CENTER LEARNING & ADAPTIVE SYSTEMS GROUP Sample Efficient Model-Based Reinforcement Learning — Grade: 5.75/6.0 (view here) ‣ Using model-based RL for efficient policy learning on a Boston Dynamics Spot quadruped with JAX, PyTorch and C++ ‣ Learning robot dynamics from data by combining Bayesian Neural Networks with physical priors ‣ Supervised by Jin Cheng, Chenhao Li, Bhavya Sukhija, Lenart Treven, Prof. Stelian Coros, and Prof. Andreas Krause	Zurich, Switzerland Jun. 2024 - Dec. 2024
SEMESTER THESIS — NETWORKED SYSTEMS GROUP Machine Learning Based Video Streaming Algorithms — Grade: 6.0/6.0 (view here) ‣ Researched and evaluated state-of-the-art ML-based video streaming algorithms ‣ Developed Python-based network-data collection and benchmarking platform using Docker and AWS/Azure SDKs ‣ Supervised by Dr. Alexander Dietmüller and Prof. Laurent Vanbever	Zurich, Switzerland Sep. 2023 - Jan. 2024
BACHELOR THESIS — COMPUTATIONAL ROBOTICS LAB State Estimation for Legged Robots — Grade: 6.0/6.0 (view here) ‣ Researched, developed and implemented different state estimation approaches with C++ and a Unitree A1 quadruped ‣ Determined the optimal approach still being used in the Lab to date based on an analysis of speed and accuracy ‣ Supervised by Dr. Dongho Kang and Prof. Stelian Coros	Zurich, Switzerland Feb. 2021 - Aug. 2021

PUBLICATIONS

- [1] **Benjamin Hoffman**, Alexander Dietmüller, Ayush Mishra, and Laurent Vanbever. “Strategic Data Collection and Generalization for Learning-Based Adaptive Bitrate Control”. In: *Under Review* (2025).
- [2] **Benjamin Hoffman**, Alexander Dietmüller, Ayush Mishra, and Laurent Vanbever. “Into the Wild: Real-World Testing for ML-Based ABR”. In: *ACM PACMI* (Seoul, South Korea, 2025). [PDF]
- [3] **Benjamin Hoffman**, Alexander Dietmüller, and Laurent Vanbever. “Poster: Towards Network Model Generalization using Strategic Data Collection”. In: *ACM SIGCOMM* (Coimbra, Portugal, 2025). [PDF]
- [4] **Benjamin Hoffman**, Jin Cheng, Chenhao Li, and Stelian Coros. “Learning More With Less: Sample-Efficient Model-Based RL for Loco-Manipulation”. In: *Workshop on Resource-Rational Robot Learning at CoRL* (Seoul, South Korea, 2025). [PDF]

WORK EXPERIENCE

BAIN & COMPANY

Zurich, Switzerland

Associate Consultant Intern — full time intern (return offer received)

Feb. 2024 - Apr. 2024

- Advised 3 industry leading firms and Fortune 500 companies on performance improvement and IT strategy projects
- Conducted empirical research and analyzed large, unstructured datasets to formulate strategic recommendations

ERANEOS

Zurich, Switzerland

Data Science Intern — full time intern (return offer received)

Jun. 2023 - Dec. 2023

- Worked on 5+ projects in finance and the public sector with a focus on ML & GenAI, LLMs, and data analytics
- Developed custom LLM solutions and multi-agent systems using fine-tuned local LLMs, RAG, and OpenAI API

GÜDEL GROUP

Bern, Switzerland

Robotics and Software Engineer — full time intern and part time working student (return offer received)

Aug. 2021 - May 2023

- Machine learning, data science and software engineering for robotics and industrial analytics
- Transformed 3+ quality control procedures to ML-driven processes and rewrote the software for 4+ key applications

ETH ZURICH

Zurich, Switzerland

Teaching Assistant for Analysis III and Dimensioning I — part time

Sep. 2020 - Jan. 2023

- Prepared and led weekly exercise sessions for 30+ students, graded assignments and exam supervision
- Elected by students to lead a two-day exam preparation course with 30+ participants for Analysis III

EXTRACURRICULAR EXPERIENCE

UN-ETH STUDENT TEAM

Zurich, Switzerland

Project Lead

Sep. 2025 - present

- Building a voting-data processing pipeline and a user-facing visualization interface for all 3 main UN bodies
- In charge of project management, leading a team of 5 software engineers, coordinating with UN stakeholders

GLP SIGGENTHAL

Obersiggenthal, Switzerland

Board and Committee Member

Aug. 2021 - present

- In charge of communications, as well as planning and implementing campaigns and initiatives on a local level
- Member of the advisory body of our municipality on topics of nature and environmental conservation

VEREIN LEGAL HELP (NGO)

Zurich, Switzerland

Board Member and Head of IT

Sep. 2022 - Sep. 2025

- In charge of strategic initiatives, maintaining long-term growth, as well as all technical and software related projects
- Led an organization-wide (100+ users) software transition project resulting in savings of 5000+ CHF annually

ANALYTICS CLUB AT ETH

Zurich, Switzerland

Quants at ETH Member

Feb. 2023 - Dec. 2024

- Growing the quant community at ETH, organizing events of 70+ participants at ETH with leading quant firms

ETH ENTREPRENEUR CLUB

Zurich, Switzerland

Finance Committee Member

Sep. 2020 - Feb. 2022

- Led a project to automate reimbursements, invoicing, and accounting, increasing process efficiency by over 300%
- Acted as the principal financial point of contact for leading startups, 50+ members, and major corporate partners

SKILLS & CERTIFICATIONS

PROGRAMMING

Python, C++, MATLAB, JavaScript, SQL

FRAMEWORKS

PyTorch, JAX, Docker, Pandas, NumPy, SciPy, scikit-learn, OpenAI Gym & API, LangChain, Git, AWS, Azure
NVIDIA Isaac Sim, Boston Dynamics, Unitree, OpenCV

LANGUAGES

English (Native), German (Native), Swiss German (Native), French (B2)

CERTIFICATIONS

J&S Coach Tennis & Snowboard, Certificate in Portfolio and Risk Management (University of Geneva, 2023)

INTERESTS

SPORTS

Competitive tennis player (coach and team captain), snowboarding, running, strong interest in health and fitness

OTHER INTERESTS

Public Policy, law and economics, global and local politics, nature and the environment